



Using state-space shortest-path heuristics to solve the long-haul point-to-point vehicle routing and driver scheduling problem subject to hours-of-service regulatory constraints

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Received: 24 February 2021 / Revised: 17 September 2021 / Accepted: 15 December 2021 /

Published online: 10 January 2022

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Abstract

This paper addresses the vehicle routing and driver scheduling problem of finding a low cost route and stoppage schedule for long-haul point-to-point full-load trips with intermediate stops due to refueling needs and driver hours-of-service regulatory restrictions. This is an important problem for long-haul truck drivers because in practice regulatory driving limits often do not coincide with availability of stoppage alternatives for quick rest, for meal, for overnight, or for weekly downtime required stops. The paper presents a methodology and algorithm to pick routes that optimize stoppages within the HOS constraints, an important factor of both highway safety and driver productivity. A solution for this variant of the vehicle routing and truck driver scheduling problem (VRTDS-HOS) that is fast enough to potentially be used in real time is proposed by modeling possible stoppage configurations as nodes in an iteratively built multi-dimensional state-space graph and by using heuristics to decrease processing time when searching for the lowest-cost path in that graph. Individual nodes in the graph are characterized by spatial, temporal, and stoppage attributes, and are expanded sequentially to search for low-cost paths between the origin and the

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destination. Within this multi-dimensional state-space graph, the paper proposes two heuristics applied to a shortest-path algorithmic solution based on the A^* algorithm to increase processing speed enough to potentially permit real-time usage. An illustrative application to Brazilian regulations is provided. Results were successful and are reported together with sensitivity analyses comparing alternative routes and different heuristics processing speeds.

Keywords Vehicle routing and driver scheduling · Point-to-point long-haul full-load trip optimization · Driver hours of service regulatory constraints · Heuristic state-space graph search · A^* algorithm

1 Introduction

A significant portion of long-haul point-to-point overland commercial freight is accomplished by trucking, with drivers sometimes traveling continental distances and spending days on the road to deliver full loads to one destination. In recent decades most countries have implemented regulations to promote public highway safety through improved working conditions for professional truck drivers, such as limiting continuous hours at the wheel and establishing upper bounds in daily and weekly driver engagement times. Although necessary for safety reasons, these regulations usually include mandated driving interruptions of different durations (e.g., for rest, for meal, for overnight, for weekly break) besides other limitations, and impose constraints that influence the choice of itinerary given availability of support infrastructure along different routes, in an hours-of-service variant of the VRP that has generated academic attention as indicated by recent surveys (Schiffer et al. 2019; Vidal et al. 2019). The implementation of regulatory frameworks must be carefully thought through as mandated hourly driving restrictions coupled with refueling, resting, and eating availability options along alternative routes can materially impact itineraries and travel times, affecting the economics of long-haul transportation in different ways for independent drivers and trucking companies as examined in Mayerle et al. (2020). Unlike policy analysis, in-vehicle trucking route guidance systems are time-sensitive and ideally should provide quick real-time responses taking into consideration stoppage infrastructure alternatives along different routes.

This paper addresses the long-haul point-to-point variant of the vehicle routing and truck driver scheduling problem with intermediate stops (VRDSP-IS) subject to driver hours-of-service regulatory restrictions (VRDSP-HOS), i.e., jointly considering routing and stoppage scheduling alternatives and constraints. In this approach, the tradeoffs between routing costs and required stop scheduling costs inherent to the VRDSP-HOS, and stoppage configurations are modeled as a graph in a multi-dimensional state-space where individual nodes with spatial, temporal, stoppage category, and other defining parameters are expanded sequentially until the least-cost path between the origin and the destination is found. It is a multi-attribute vehicle routing problem (MAVRP) due to its unique driver and vehicle operation rules and varying decision contexts—according to Vidal et al. (2013) the use of heuristics and meta-heuristics in the academic literature to solve MAVRPs numbers in the thousands due to the intense tailoring that is

often necessary to address specific problems so the use of solution estimates is especially appropriate given the combinatorial nature of alternatives. Within the state-space graph model, the paper proposes two heuristics applied to a shortest-path algorithmic solution based on the A^* algorithm proposed by Hart et al. (1968) to lower processing time and potentially permit real-time usage. In doing so, this paper rekindles one of the original artificial intelligence algorithms as a source of inspiration to address MAVRP problems.

The combination of infrastructure constraints that affect driver resting and vehicle refueling options with regulatory requirements that establish rigid hours-of-service time windows renders the long-haul point-to-point VRDSP-HOS with hours-of-service regulatory constraints as important in countries with vast area and sparsely populated regions such as Brazil, Canada, Russia, and Australia. However, it is also relevant in more densely populated areas with many routing alternatives, such as the European Union, first addressed by Goel (2009) who, as described in the next section's literature review, has examined different regulatory environments, as well as the eastern part of the United States for which Rancourt et al. (2013) proposed a multiple time-window solution. This paper adds the example of Brazil, a country that in March 2015 passed trucking Law n° 13,103 (BRASIL 2015) establishing maximum uninterrupted service hours and minimum required rest periods for truck drivers at various levels (i.e., meals, overnight, and for weekly rest), as well as other restrictions. The law's constraints together with the sparse distribution of rest options in some parts of Brazil underline the importance of developing algorithms to enhance freight transportation efficiency and highway safety.

1.1 Contributions to the literature and structure of the paper

This research contributes to the VRDSP-HOS literature by addressing an issue raised by Vidal et al. (2019) in their guide to VRP variants, namely that many stoppage decisions reside with the drivers and are made in real-time. The authors specifically mention that there is a thin line between regulatory aspects that can be optimized ex-ante and those that should be used as a recourse when drivers face unforeseen events on the road, therefore implicitly underlining the need for methods that reach solutions faster so they can be used in in-vehicle route guidance systems. While managers can use this research to *a priori* obtain a blueprint plan to minimize costs in any given long-haul trucking route, it also can provide flexibility to drivers when travel conditions suddenly change due to unexpected traffic or weather impediments, for example. This flexibility derives from the model's capacity to determine alternative routes that take into account availability and costs of stoppage options. Another contribution of this research is the state-space modeling itself, as nowhere in the literature did we find a treatment of the long-haul VRTDS-HOS problem in which times of arrival and departure in every stop along the way were treated as continuous variables, therefore allowing for an examination of route alternatives within a shortest-path framework with both temporal and spatial characteristics besides stoppage attributes.

Within the framework just described, this paper contributes to the VRDSP-HOS literature in three specific ways, in addition to the broad conceptual methodology

already presented by Mayerle et al. (2020). The three contributions are as follows. First, this research uses an adapted label-setting A^* algorithm to find that shortest path in the iteratively built multidimensional state-space graph. Second, the paper proposes two heuristics, one of which is innovative in that it uses cost data obtained as the trip evolves, i.e., it “learns” from information acquired during the travel time from the origin through each node expansion, to narrow the number of feasible successor nodes and therefore lower processing time even further. Third, this paper adds Brazil to the several country truck driving regulatory frameworks that have been examined in the DSP-HOS and the VRDSP-HOS literatures as described in the next section, and in doing so illustrates the use of the algorithm within the current Brazilian context.

This section introduces the paper with a brief overview of the problem while the next section provides a literature review. The third section explains the mathematical formulation and introduces the state-space graph that models this combinatorial optimization problem, describing the multi-dimensional nature of the nodes and the sequential process of node expansion to construct the graph. The fourth section covers the process of searching for the shortest path in this graph by proposing two heuristics to lower processing time and by describing the A^* algorithm adaptation. The fifth section illustrates the use of the algorithm in Brazil and compares processing times using the two heuristics to that with none (i.e., with the Dijkstra algorithm adapted to the state-space graph). The final section concludes.

2 Background literature overview

The vehicle routing problem is essentially a spatial problem, while the driver scheduling problem involves spatial and temporal characteristics, and the hours-of-service component adds yet another set of temporal attributes to each stoppage decision, whether it is required by regulations or by operational needs. Given its many different variations, the vehicle routing problem (VRP) has been among the most studied applications in combinatorial optimization and we begin this literature review by positioning the long-haul point-to-point vehicle routing and driver scheduling problem subject to hours-of-service regulatory constraints within the classifications of several recent academic surveys. Laporte (2009) celebrated fifty years of the VRP by providing a top-down framework to classify the proposed solution algorithms as exact, heuristic (as is the case of this research), or metaheuristic. Coelho et al. (2014) suggested a comprehensive categorization when surveying the inventory routing variant of the problem according to which this research addresses one vehicle in a point-to-point (one-to-one) structure, with a finite time-horizon, and direct routing. Following Schiffer et al. (2019)’s categorization of vehicle and location routing, which includes hours-of-service driver scheduling, this research optimizes route duration with intermediate stops for refueling as well as for idling due to regulatory requirements for required rest periods and breaks. Finally, in Vidal et al. (2019)’s examination of VRP problem variants, the problem addressed in this research is classified under working hours regulations in the driver and vehicle specificities category. We now examine the literature on the impact of regulatory hours-of-service (HOS) constraints on the VRDSP, with a brief summary of papers examining only driver scheduling followed

by a more detailed inspection of papers that also include routing optimization. We then summarize the point-to-point shortest-path literature focusing on heuristics used to speed up processing time. Lastly, to conclude this overview of the literature we turn to the few prior applications the A^* algorithm within the VRP context.

Much research has focused on the driver scheduling problem with HOS requirements either for single or multiple vehicles (fleet management) and either for point-to-point travel or multiple required stops for replenishment and/or delivery. Savelsbergh and Sol (1998) were the first to allow for meal and end-of-day breaks for pickup and delivery scheduling using a branch-and-price algorithm and column generation. Campbell and Savelsbergh (2004) included driver shift time limits as a capacity constraint and addressed this and other real-life complicating factors through insertion heuristics. Ceselli et al. (2009) included regulatory upper limits on driving times and required rest periods in a rich vehicle-routing column generation algorithm to provide daily plans for a heterogeneous fleet in a many-to-many structure with time windows. Archetti and Savelsbergh (2009) addressed the driver scheduling problem under U.S. HOS regulations for full-load driver scheduling with more than one sequential pick-up and delivery with dispatching at the origin, and provided a polynomial time algorithm to schedule stops minimizing deadhead moving costs. This was followed by several papers solving the long-haul truck driver scheduling problem under different regulatory regimes, subject to different specific operational attributes, and/or using different algorithmic solutions (Goel 2012a, b; Goel and Vidal 2014; Goel et al. 2012; Goel and Kok 2012; Goel et al. 2012). These papers did not address the VRDSP-HOS, i.e., the problem of simultaneously optimizing the travel route together with driver scheduling by examining tradeoffs between displacement and rest cost alternatives along different routes, which has been the focus of more recent research.

Xu et al. (2003) were the first to consider actual U.S. driver working rules in optimizing vehicle routing and driver scheduling and addressed the problem by using set partitioning and solving its consequent linear relaxation through a column generation procedure. Goel (2009) proposed a vehicle routing and driver scheduling algorithm using large neighborhood search techniques in the first paper specifically addressing the European Union HOS regulations established in 2007. Prescott-Gagnon et al. (2010) applied a tabu search algorithm to generate routes by inserting customers, Kok et al. (2010) used a dynamic programming heuristic for the VRP with time windows, and Drexler and Prescott-Gagnon (2010) proposed the use of shortest path labelling algorithms with resource constraints, whether exact or using heuristics. Rancourt et al. (2013) provided a comprehensive fleet analysis of long-haul vehicle routing and driver scheduling under U.S. HOS regulations and solved it using tabu search heuristics. Goel and Vidal (2014) developed an algorithm combining metaheuristics, local search and a forward label procedure to compare the impact on carrier fleets of HOS regulations in the European Union, Australia, the United States, and Canada. Goel and Irnich (2017) developed the first exact algorithm to solve the multiple vehicle with multiple stops routing and driver scheduling problem by extending a time-windows version to include HOS regulations and applying a branch-and-price algorithm. Koç et al. (2017) solved the VRDSP with different types of idling options taking into account hours-of-service regulations in the U.S., which distinguish non-driving times between on-duty and off-duty depending on whether the driver is performing obligations other

than driving or on break or rest, and developed an algorithm combining mixed integer programming with an adaptive large neighborhood search. Goel (2018) provided a detailed analysis of HOS regulations in the European Union and specifically examined the impact of including night work provisions on route and schedule optimization. The tradeoffs between asset (truck) utilization and the choice between manning long-haul routes with single or team driving in the European Union were addressed by Goel et al. (2019). Alcaraz et al. (2019) examined a multi-attribute long haul many-to-many VRP with HOS regulatory constraints, heterogeneous fleet, and last-mile outsourcing with both construction heuristics and improvement heuristics.

The academic literature on different variations of shortest-path approaches to network problems, whether exact or using heuristics, is extensive. Examples of such distinct approaches are the use of penalty functions for resource-constrained problems (Avella et al. 2002); an exact algorithm that can be used heuristically by truncating execution to solve the robust shortest-path problem with interval data (Montemanni and Gambardella 2004); an algorithm to solve the piecewise continuous shortest-path problem in time-dependent networks (Dell'Amico et al. 2008); a highway hierarchy-based shortest-path heuristic algorithm for static large-scale networks with periodic traffic information queries evaluated and updated dynamically (Nannicini et al. 2010); the use of min-max methods, helpful to identify best worst-case alternatives in multi-criteria problems, both exact (Murthy and Her 1992); and heuristic (Bruni and Guerriero 2010), the latter of which addressed the robust shortest path problem; and the use of linear-programming-based branch-and-bound methods to solve resource-constrained problems (Horváth and Kis 2016). As exemplified in the previous paragraph for the specific case of HOS applications, VRP research has also relied extensively on shortest-path approaches, most recently within the contexts of the pollution-routing problem in which the departure time and speed are optimized (Dabia et al. 2017 and Franceschetti et al. 2018); of planning e-vehicle displacements given necessary charging stops (Baum et al. 2019); and of ensuring rest and overnight parking availability given an ordered set of client stops (Vital and Ioannou 2021), for example.

The A^* algorithm, which is based on a best-first search with the addition of formal heuristics, was introduced over fifty years ago by Hart et al. (1968), but unlike the original label-setting shortest-path algorithm (Dijkstra 1959), was not developed for Operations Research applications, but rather was one of the first results in the field of artificial intelligence. However, despite its explicit use of distance estimates for robot movements (Nilsson 1971), with a bi-directional variant suggested shortly after its development (Pohl 1971), the A^* algorithm did not get very much immediate attention from the OR community (Pijls 2007). Indeed, although the A^* is a label-setting algorithm that essentially uses heuristics to improve over Dijkstra's by reducing the number of expanded nodes in a goal-directed way, initially its application to real-world transportation networks was sparse. Early published research on other innovative heuristics algorithms of the same class, whether unidirectional or bidirectional, are exemplified by Goldberg and Harrelson (2005) and Klunder and Post (2006). Cordeau et al. (2002) provided a concise guide to VRP heuristic methods stressing that accuracy, speed, simplicity, and flexibility are the main attributes of good heuristics. In their state-of-the-art survey of heuristic shortest path algorithms for transportation, Fu et al. (2006) stressed the then-growing importance of shortest-path heuristics for

applications requiring quick and accurate response such as in-vehicle route guidance systems and real-time automated vehicle dispatching systems. They classify heuristic methods that limit the search area into **branch pruning techniques**, in which the nodes with a low probability of being on the shortest path are pruned, and the **A* algorithm**, which keeps those nodes but assigns them a higher weight or penalty than more promising ones. They listed studies documenting the performance of the A* algorithm in Euclidean networks and concluded by saying that further research to realize the full potential of heuristic search algorithms was needed. Much of the ensuing research on shortest-path heuristics reported in the literature remained focused on variations of the original A* algorithm as directly applied to physical road grids. Several examples illustrate this research, such as static shortest path multi-objective optimization "NAMOA" (Madow and De La Cruz 2008); bi-directional search in time-dependent road networks (Nannicini et al. 2012); dynamic shortest path problems using a decremental approach to speed up processing time when necessary (Ardakani and Tavana 2015); and shortest path problems with learning effects (Wang et al. 2016) for example. Just as relevant to the research presented herein are previous examples of applying the A* algorithm in multi-dimensional state-space approaches with significant modeling complexity such as optimizing tower spotting in transmission line construction (de Figueiredo and Gonzaga 2003) and optimizing spreading and cutting for garment manufacturing (Nascimento et al. 2010). Indeed, one objective of the research herein is to examine the possibilities inherent in expanding the optimization state-space in VRP problems to include several new dimensions beyond the spatial and temporal ones, as described in Sect. 3.

3 Problem formulation and state-space modeling

The long-haul point-to-point vehicle routing and driver scheduling problem subject to hours-of-service regulatory constraints is now formally defined and, because of its exponential nature, it is then modeled as a shortest-path problem in a graph in which each node encapsulates spatial, temporal, and category attributes of potential stoppage configurations. The spatial constraints are similar to those of traditional routing problems, but adding HOS regulatory restrictions (e.g., length of uninterrupted driving time, meal duration, maximum allowed daily driving times, minimum overnight rest duration, and allowed time intervals for daily meals) changes the number of feasible options and significantly increases the magnitude of this combinatorial problem, as described in Mayerle et al. (2020).

3.1 Decision variables, objective function, and constraints

This section characterizes the problem in mathematical terms.

3.1.1 Decision variables

y_{ij}^k is a decision variable indicating whether travel between a given pair of locations i and j occurs on the k^{th} day with travel time $t(i, j)$ and cost $c(i, j)$, where y_{ij}^k is one if the driver decides to go from i to j on day k , and zero otherwise.

$W_{ip}^k \in \{0, 1\}$ is a decision variable indicating whether location i is used for a stop of type $p \in P$ on day k , where $p \in \{1, 2, 3, 4\}$ represent different stop modes, respectively rest, meal, overnight, and weekend downtime, where W_{ip}^k is one if the driver decides to stop at location i for reason p on day k , and zero otherwise.

The number of driving hours in day k , the continuous decision variable h_k , is a direct result of the decisions above.

The times at which the driver enters and leaves stoppage location i are two continuous decision variables, represented by:

a_i arrival time in location i ;
 d_i departure time from location i .

3.1.2 Objective function

The objective of the problem is to minimize the costs incurred during the trip. This formulation includes four categories of costs:

The first cost category of this objective function comprises all travel displacement costs excluding driver labor, expressed as the sum of displacement costs $c(i, j)$ associated with each trip segment.

The second cost category comprises all types of stoppage costs and is the sum of all costs incurred during stops for rest, meals, overnight and weekend downtime. A cost $d(i, p)$ is associated with each stop type p and location i .

The third cost category, denoted by $\eta(h_k)$, represents driver labor costs. These costs are not linear. For example, if driving time is less than the normal daily work time in a given workday the driver still receives a base salary. Overtime pay, when necessary, also provides a discontinuity. This cost category is a function of h_k , the number of hours driving during the k^{th} day.

A fourth cost component, $\phi(d_s, a_t)$, is the opportunity cost of having a truck in use, the forgone contribution to operating profits of a transportation asset (in this case the truck) not being utilized in another alternative transportation task. This cost is a function of the time each truck is not in use, namely the difference between the departure time from a stoppage and the respective arrival.

Equation (1) is the problem's cost minimization objective function.

$$\min Z = \sum_k \sum_{(i,j) \in A} c(i, j) y_{ij}^k + \sum_k \sum_{p \in P} \sum_{i \in N} d(i, p) W_{ip}^k + \sum_k \eta(h_k) + \phi(d_s, a_t) \quad (1)$$

The problem is characterized by physical constraints, temporal constraints, and constraints to ensure variable type.

3.1.3 Physical constraints

Physical constraints relate to flow conservation, existence of an origin and a final destination, and the requirement that there not be more than one stop at any given location.

Equation (2) below ensures flow conservation given a set N of all possible allowed stop locations offering either rest, meal, and/or overnight facilities along possible routes. Equations (3) and (4) characterize the origin and the destination respectively denoted by $s \in N$ and $t \in N$.

$$\sum_k \sum_{j|(i,j) \in A} y_{ji}^k = \sum_k \sum_{j|(i,j) \in A} y_{ij}^k \quad \forall i \in N - \{s, t\} \quad (2)$$

$$\sum_i y_{si}^1 = 1 \quad (3)$$

$$\sum_k \sum_{i|(i,t) \in A} y_{it}^k = 1 \quad (4)$$

Equation (5) ensures that $W_{ip}^k = 1$ if a stop of type p occurs at into location i on the k^{th} day. Otherwise the variable is 0.

$$\sum_{j|(j,i) \in A} y_{ji}^k = \sum_{p \in P} W_{ip}^k \quad \forall i \in N - \{s, t\} \forall k \quad (5)$$

Equation (6) ensures that any location either has one stop or none.

$$\sum_k \sum_{p \in P} W_{ip}^k \leq 1 \quad \forall i \in N \quad (6)$$

3.1.4 Temporal constraints

Depending on stoppage category attributes, there are several HOS requirements, which are the temporal constraints of the formulation.

Equation (7) accumulates travel times on the k^{th} day.

$$h_k = \sum_{(i,j) \in A} t(i, j) y_{ij}^k \quad \forall k \quad (7)$$

Equation (8) limits the accumulated daily travel time to the maximum permitted daily driving respectively in normal and overtime labor hour conditions ($L_{nor} + L_{ovr}$).

$$h_k \leq L_{nor} + L_{ovr} \quad \forall k \quad (8)$$

Equation (9) ensures that the departure time from each stoppage location is in compliance with the required minimum length of time $T(p)$ associated with the respective stop type p .

$$d_i \geq a_i + \sum_k \sum_p T(p) W_{ip}^k \quad \forall i \in N - \{s, t\} \quad (9)$$

Equation (10) sets the departure time from the origin.

$$d_s \geq W_1^{start(day)} \quad (10)$$

Equation (11) establishes the arrival time at each stop as the sum of the departure time from the previous stop and the elapsed travel time to the next stop, where A is the set of trip segments linking any two locations i, j such that travel time along the respective connection is lower than the maximum allowed uninterrupted driving time, denoted by L_{drv} .

$$a_j \geq d_i + \sum_k t(i, j) y_{ij}^k \quad \forall (i, j) \in A \quad (11)$$

Equation (12) requires that the departure time after an overnight on the k^{th} day be in compliance with minimum overnight period regulations (M is a penalty allowing for the optional disabling of the equation).

$$d_i \geq W_{k+1}^{start(day)} + M(W_{i3}^k - 1) \quad \forall i \in N \quad \forall k \quad (12)$$

If the main meal of the day must occur within a specific time window $[W_k^{start(meal)}, W_k^{end(meal)}]$, Eqs. (13) and (14) ensure that arrival time intervals at a location $i \in N$ for a meal on the k^{th} day will occur within that time interval (M is a penalty allowing for the optional disabling of the respective equation).

$$a_i \geq W_k^{start(meal)} + M(W_{i2}^k - 1) \quad \forall i \in N \quad \forall k \quad (13)$$

$$a_i \geq W_k^{end(meal)} + M(1 - W_{i2}^k) \quad \forall i \in N \quad \forall k \quad (14)$$

Equations (15) and (16), ensure at least one stop for a meal and for an overnight each day (M is a penalty allowing for the optional disabling of the respective equation).

$$M \sum_{i \in N} W_{i2}^k \leq h_k \quad \forall k \quad (15)$$

$$M \sum_{i \in N} W_{i3}^k \geq h_k \quad \forall k \quad (16)$$

3.1.5 Variable type constraints

Equations (17) to (21) identify the types of variables (i.e., whether binary, integer, or continuous).

$$y_{ij}^k \in \{0, 1\} \quad \forall (i, j) \in A \quad \forall k \quad (17)$$

$$W_{ip}^k \in \{0, 1\} \quad \forall i \in N \quad \forall p \quad \forall k \quad (18)$$

$$a_i \geq 0 \quad \forall i \in N \quad (19)$$

$$d_i \geq 0 \quad \forall i \in N \quad (20)$$

$$h_k \geq 0 \quad \forall i \in N \quad (21)$$

3.2 The state-space graph model

The problem is now modeled as a search problem in a state-space directed graph in which each node represents a unique multi-dimensional state that is described by spatial, temporal, and stoppage attribute components, i.e., in which each node is a unique stoppage configuration that is fully described by a set of nine parameters. Arcs between nodes represent transitions from one such state to another. The graph in its entirety is not known explicitly beforehand because evolving conditions can lead to different feasible stoppage configurations. As time advances only certain states (or nodes) within the state-space can be reached from a given node without violating any constraints. Using the original A^* terminology (Hart et al. 1968), the expansion of a node is the process of determining the set of feasible successors to any given node. Given any stoppage configuration or state represented as a node in the graph, its expansion is the identification of subsequent viable stops, with each such stoppage configuration labeled a successor to that node. The process of building the directed state-space graph consists of successive node expansions following a label-setting shortest path algorithm beginning at the origin, in which each node is a configuration of stoppages and each arc connects two different adjacent stoppage configurations. This graph is characterized by a set of nodes, by a set of arcs that are ordered pairs of nodes, and by costs associated to arcs, which are now made explicit.

3.2.1 Node

A node in the state-space graph, identified by x_n , is defined as an nine-tuple containing all the information necessary to identify a stoppage configuration. This information completely defines the state, and includes the permitted location ID; the type of stop (rest, meal, overnight, or weekly rest); the travel day; the driving time durations since the previous stop, since the morning, and since the beginning of the week; and the arrival and departure times, in addition to an indicator of whether the required main meal of the day has happened or not. A node is represented by:

$$x_n = (i_n, p_n, k_n, h_n^1, h_n^2, h_n^3, a_n, d_n, \delta_n) \quad (22)$$

in which:

i_n is the stoppage location ID associated with node x_n ;

p_n is the type of stop associated to node x_n , where the value of $p_n \in \{1, 2, 3, 4\}$ corresponds respectively to a short rest stop, to a meal, to an overnight stay, or to the required weekly rest;

k_n is the current travel day in node x_n , counted sequentially beginning in the departure from the origin (the first day corresponds to $k_n = 1$);

h_n^1 is the driving time since previous rest stop;

h_n^2 is the driving time that day;

h_n^3 is the driving time that week;

a_n is the arrival time at node x_n ;

d_n is the departure time from node x_n ;

δ_n ($\delta_n = 1$) if the main meal of the day has happened or ($\delta_n = 0$) if it has not.

The spatial location of the stop is only one of nine different parameters or dimensions defining the node or state because the state of the node is defined by stoppage attributes (discrete) and temporal markers (continuous).

3.2.2 Arc

If node x_m precedes node x_n , the latter is one of the nodes obtained by expanding node x_m and these two nodes are linked by an arc. Let $t(i_m, i_n, d_m)$ be the travel time between locations i_m and i_n given departure time d_m , and consider that after stopping at i_m the driver will make a subsequent stop at i_n and that this stop will be of type $p_n \in \{1, 2, 3, 4\}$. The following equations define the additional seven parameters characterizing the successor node:

$$k_n = \begin{cases} k_m, & \text{if } p_m \in \{1, 2\}, \\ k_m + 1 & \text{if } p_m = 3, \\ k_m + 2 & \text{if } p_m = 4 \end{cases} \quad (23)$$

$$h_n^1 = t(i_m, i_n, d_m) \quad (24)$$

$$h_n^2 = \begin{cases} h_m^2 + t(i_m, i_n, d_m) & \text{if } p_m \in \{1, 2\}, \\ t(i_m, i_n, d_m) & \text{if } p_m \in \{3, 4\} \end{cases} \quad (25)$$

$$h_n^3 = \begin{cases} h_m^3 + t(i_m, i_n, d_m) & \text{if } p_m \in \{1, 2, 3\}, \\ t(i_m, i_n, d_m) & \text{if } p_m \in \{4\} \end{cases} \quad (26)$$

$$a_n = d_m + t(i_m, i_n, d_m) \quad (27)$$

$$d_n \geq \begin{cases} a_n + T_1 & \text{if } p_n = 1, \\ \max[a_n, W_{k_n}^{\text{start(meal)}}] + T_2 & \text{if } p_n = 2, \\ \max[a_n + T_3, W_{k_n+1}^{\text{start(day)}}] & \text{if } p_n = 3, \\ \max[a_n + T_4, W_{k_n+2}^{\text{start(day)}}] & \text{if } p_n = 4 \end{cases} \quad (28)$$

$$\delta_n = \begin{cases} 0 & \text{if } p_n \in \{3, 4\}, \\ \delta_m & \text{if } p_n \in \{1\}, \\ 1 & \text{if } p_n \in \{2\} \end{cases} \quad (29)$$

where T_1 , T_2 , T_3 and T_4 are a required minimum length of time to a rest stop, a meal, an overnight stay, and a weekly rest.

Equations (23) determine the day in which a stop described by node x_n occurs, counting from the beginning of the route depending on whether the previous stop, described by node x_m , was a meal stop or a rest stop, in which case the next stop, described by configuration x_n will occur on the same day, or an overnight stop, in which case the next stop will occur the next day. If x_m was a weekly rest stop, the subsequent stop will occur only after the driver remains at that stop at least for the required weekly

downtime, usually one whole day, so two days are added. Equation (24) requires that uninterrupted driving time be the elapsed time between two consecutive stop locations. Equations (25) accumulate the elapsed time in the current workday, expressing it as the sum of the accumulated driving time in the day and the driving time between location stops if the previous stop was for rest or meal. If the previous stop was an overnight or weekly rest stop, the daily workday time count is reinitiated. Equations (26) are analogous, but refer to elapsed time in the current workweek, a count which reinitiates after a weekly downtime stop. Equation (27) determines the arrival time associated with the stoppage configuration described by node x_n , considering that the driver left the previous stop location i_m at the time d_m and took $t(i_m, i_n, d_m)$ in transit until arriving at location i_n . Equations (28) are used to determine if the departure time from the stoppage configuration described by node x_n is feasible, depending on the respective stop type and arrival times. If node x_n refers to a rest stop ($p_n = 1$), departure cannot occur earlier than T_1 units of time after arrival. If it refers to a meal stop ($p_n = 2$), departure cannot occur earlier than an elapsed time T_2 for the meal. In addition, the earliest beginning time for meals at the k_n^{th} day cannot occur before $W_{k_n}^{start(meal)}$. In case x_n is an overnight stop ($p_n = 3$), the minimum rest period is given by T_3 , and the following workday (i.e., driving) cannot begin before time $W_{k_n+1}^{start(day)}$. If the stop is for the weekly downtime rest ($p_n = 4$), the total minimum stoppage time is T_4 , and the following workweek (i.e., driving) cannot begin before time $W_{k_n+2}^{start(day)}$. Finally, Equations (29) are used to update δ_n , the parameter that signals whether the driver has had the main meal of the day.

The existence of an arc (x_m, x_n) is conditioned to h_n^1 , h_n^2 and h_n^3 satisfying the driving time requirements set by regulations, including labor agreements:

$$h_n^1 \leq L_{drv} \quad (30)$$

$$h_n^2 \leq L_{nor} + L_{ovr} \quad (31)$$

$$h_n^3 \leq L_{week} \quad (32)$$

in which L_{drv} , $L_{nor} + L_{ovr}$ and L_{week} are, respectively: the maximum allowed continuous driving period (i.e., without any rest stops); the maximum allowed driving time in a workday (normal plus extra labor hour); and the maximum allowed driving time in a workweek. In addition, the time of the main meal in the k_n^{th} workday needs to occur within the $[W_{k_n}^{start(meal)}, W_{k_n}^{end(meal)}]$ time window so a stop configuration with $\delta_n = 0$ on the k_n^{th} workday is only feasible if arrival occurs before $W_{k_n}^{end(meal)}$, that is:

$$a_n \leq W_{k_n}^{end(meal)} \quad \text{if} \quad \delta_n = 0 \quad (33)$$

3.2.3 Node expansion

Equations (23) to (33) determine the feasibility of a transition from one stoppage configuration (e.g., node x_m) to another (e.g., node x_n). Any two stoppage configurations, i.e., any two nodes in the graph described by $x_m = (i_m, p_m, k_m, h_m^1, h_m^2, h_m^3, a_m, d_m, \delta_m)$ and $x_n = (i_n, p_n, k_n, h_n^1, h_n^2, h_n^3, a_n, d_n, \delta_n)$, will be connected by an arc (x_m, x_n) if

and only if it is feasible for the driver to transition (relocate) from the state represented by the stoppage configuration described in node x_m , to the state represented by the stoppage configuration described in node x_n , and there have a stop of type p_n . Consequently, from any node x_m a set of feasible states, i.e., of feasible stoppage configurations represented by adjacent nodes $\{x_n, x_{(n+1)}, x_{(n+2)}, \dots\}$, is generated by expanding node x_m . Given the continuity of time-dependent elements or parameters that comprise a node's nine-tuple (such as driving times as well as arrival and departure times), the state-space graph has an unlimited number of successor nodes. In order to limit the size of the state-space representation (i.e., the size of the graph), node pruning can be used to generate only the most promising successor nodes during an expansion.

Node expansion, or the set of stoppage configurations that are feasibly reached from a given stoppage configuration x_m , is thus obtained by: (a) examining the Cartesian product of all possible stoppage types $p \in \{1, 2, 3, 4\}$ and locations i_n which can be reached from i_m in a time not greater than the regulated maximum uninterrupted driving time (L_{drv}), considering as departure time only the minimum feasible value d_m ; and b) eliminating options that do not satisfy one or more of Equations (23) through (33), i.e., eliminating all infeasible candidates. The option of considering only the minimum feasible departure time stems from the fact that less stoppage time is better than more because there are significant costs that increase with the total duration of the trip, such as the opportunity cost, cost of capital applied to fixed assets, and cost of parking, among others. This process of expanding a node x_m can be represented by a successor operator Γ applied to a node x_m , building out the graph through sequential applications of Γ :

$$\Gamma(x_m) = \{x_n, x_{n+1}, x_{n+2}, \dots\} \quad (34)$$

3.2.4 Cost of an arc

The cost associated with each arc (x_m, x_n) includes the cost of physical movement (displacement) from stoppage location i_m to stoppage location i_n and the cost incurred for services in a stop of type $p_n \in \{1, 2, 3, 4\}$ at location i_n , among others. The following are the cost components: driver compensation including benefits, which is a function of hours on the job, hours of overtime driving, social security and other taxes, and benefits such as health and life insurance and other components; truck displacement costs, including fuel, other supplies such as lubricants, and maintenance expenses including tires, which are usually a function of the time and/or distance covered; vehicle fixed costs, such as depreciation, the cost of immobilized capital, taxes, insurance, which usually are proportional to the total elapsed time of each activity or trip leg; costs of services incurred in stoppage locations which are charged per unit of time (parking expenditures, etc.), i.e. costs which are a function of the elapsed time incurred; costs of services incurred in stoppage locations which are not charged per unit of time (meals, overnight stays, etc.); the opportunity cost for the allocation of the vehicle to this trip, which is proportional to the total elapsed travel time and represents the expected value of any lost revenue incurred for not assigning the particular truck to another job or trip; a potential penalty cost to express the need

to arrive at the location of the expanded node before the respective HOS deadline; and driver labor costs.

In summary, the cost of an arc is the sum of truck displacement costs from one stoppage location to the next, time dependent stoppage costs, non-time dependent stoppage costs, the truck's opportunity cost, a penalty if regulatory requirements are not met, and driver labor costs.

Formally, the cost $c(x_m, x_n)$ associated to an arc connecting configurations represented by nodes x_m and $x_n \in \Gamma(x_m)$, may be calculated as follows:

$$c(x_m, x_n) = c_1 t(i_m, i_n, d_m) + c_2(i_n)[d_n - a_n] + c_3(i_n, p_n) + c_4[a_n - d_m] + c_5 \max[0, a_n - \max(T_{max}, a_m)] + c_6(h_m^2, h_n^2, p_m) \quad (35)$$

in which:

c_1 is the cost of truck displacement per unit of time expressed as \$/hour;

$c_2(i_n)$ is the stoppage cost per unit of time in location i_n , expressed as \$/hour;

$c_3(i_n, p_n)$ is the cost of non-time-dependent services incurred in stoppage location i_n during a stop of type p_n ;

c_4 is, without loss of generality, the opportunity cost applied to the time incurred between departure from stoppage location i_m and arrival to stoppage location i_n and described above, which is considered to be linear in time;

c_5 is the penalty cost per hour incurred if deadline T_{max} is not respected; and

$c_6(h_m^2, h_n^2, p_m)$ is the daily labor cost which includes costs described above and is calculated as:

$$c_6(h_m^2, h_n^2, p_m) = \begin{cases} S(h_n^2) - S(h_m^2) & \text{if } p_m \in \{1, 2\}, \\ S(h_n^2) & \text{if } p_m \in \{3, 4\} \end{cases} \quad (36)$$

where $S(h)$ is the driver cost for h working hours in the respective day, given by:

$$S(h) = \begin{cases} S_{nor} L_{nor} & \text{if } h \leq L_{nor} \\ S_{nor} L_{nor} + S_{ovr}(h - L_{nor}) & \text{if } L_{nor} < h \leq L_{nor} + L_{ovr} \end{cases} \quad (37)$$

where S_{nor} and S_{ovr} correspond to the hourly labor driving costs (including benefits) in both normal and overtime and L_{nor} and L_{ovr} are respectively the maximum permitted daily normal and overtime driving times.

In Equation (35), the driver labor cost on arc (x_m, x_n) is calculated in two different ways depending on the type of stoppage configuration x_m . If stoppage configuration x_m corresponds to an overnight ($p_m = 3$) or to a weekly rest ($p_m = 4$), the labor cost on the respective arc will take into account only the driving time that day until stoppage configuration x_n , that is, h_n^2 . If configuration x_m corresponds to a rest ($p_m = 1$) or meal ($p_m = 2$) stop, labor driving costs on the respective arc through configuration x_n should be calculated based on the day's driving time, h_n^2 , less the labor driving costs through the predecessor configuration.

4 Finding the shortest path in the state-space graph with A^* heuristics

4.1 Proposed A^* heuristics in the state-space graph

The route and stoppage schedule of the lowest cost trip of a full-load point-to-point trucking delivery assignment between two locations is the shortest path of stoppage configurations obtained by sequentially expanding the state-space graph nodes beginning with the initial state associated with the origin and ending at a final state associated with arrival to the destination. This state-space graph may not be a tree because there could be several ways to reach a given stoppage configuration.

The A^* algorithm is a label-setting shortest path in the same vein as Dijkstra (1959), that associates with each node $x_n \in X$ a label $[\hat{g}(x_n), \text{pred}(x_n)]$, in which $\hat{g}(x_n)$ was the current lowest cost sequence of nodes connecting the origin x_s to node x_n , and $\text{pred}(x_n)$ was the predecessor (the preceding node) to x_n in the sequence. Hart et al. (1968) introduced a heuristic approach (the A^* algorithm) with the objective of reducing processing time while losing little or nothing in terms of optimality. A^* replaced the test for labeling a node based only on costs already incurred with a test that adds to the incurred cost an estimated cost remaining from each labeled node to the final destination.

In A^* , the expansion of a node is made based not only on the lowest cost between the origin and the respective node $x \in X$, $\hat{g}(x)$, but also on adding an estimate of the cost $\hat{h}(x)$ between this node x and the final destination x_f . Let $\hat{f}(x_m) = \hat{g}(x_m) + \hat{h}(x_m)$ be an estimate of the lowest cost path connecting the origin node to the destination node through node x_m . Instead of selecting node x with the lowest value of $\hat{g}(x)$, an alternative node $x_m \in X_T$ with the lowest value of $\hat{f}(x_m)$, i.e. $\hat{f}(x_m) = \min_{x \in X_T} \hat{f}(x)$ is selected. Nilsson (1971) demonstrated that if $\hat{f}(x)$ is an optimistic estimate for every $x \in X$, i.e., if the estimated cost undercuts the actual cost for every alternative in the complete graph, optimality of the search is guaranteed. Conversely, if $\hat{h}(x)$ does not undercut the lowest cost for every $x \in X$ in the complete graph there is no guarantee of optimality but the cost of the respective solution will not exceed the cost of the optimal solution by more than the ratio by which the estimate $\hat{h}(x)$ exceeds the exact $h(x)$.

In this paper we propose two different functions for $\hat{h}(x)$ in addition to $\hat{h}_I(x_n) = 0$, in which no information is added to the search process and the algorithm behaves like Dijkstra's search method.

The first heuristic consists of estimating the cost from any node to the destination node in the simplest possible way, namely considering that the remaining distance to be covered is a straight line. The cost estimate assumes the truck covers the distance in ideal speed conditions, without any resting stops. In this case the heuristic function becomes $\hat{h}_{II}(x_n) = \alpha \text{dist}(i_n, t)$, in which the parameter α is a constant based on the vehicle's displacement operating cost and $\text{dist}(i_n, t)$ is the straight line distance from the location associated with stop location i_n to the final destination t .

The second heuristic $\hat{h}_{III}(x_n)$ estimates the remaining displacement and rest stop costs by extrapolating the cost patterns that have been experienced up to the given

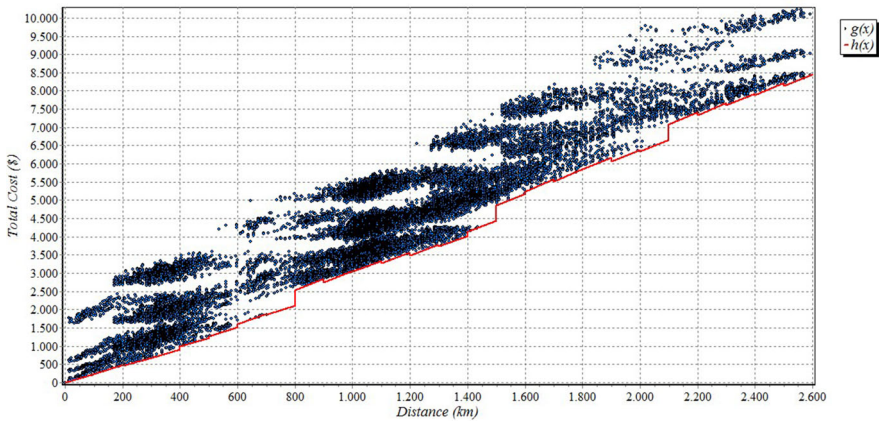


Fig. 1 Realized displacement and rest cost envelope for each interval of straight line travel in a representative case

trip stage. In other words, $\hat{h}_{III}(x_n)$ assumes that the cost patterns in the route going forward will not be inferior to the costs on the route up to that specific point, i.e., that the remainder of the route will be covered with displacement and a pattern of stoppages with costs greater or equal to those experienced up to that point. Consider that for every node already expanded, i.e., labeled as *permanent* ($x_n \in X_P$), there is a pair of values $[dist(s, i_n), \hat{g}(x_n)]$, where $dist(s, i_n)$ is the straight line distance from the beginning of the trip to location i_n and $\hat{g}(x_n)$ is the current lowest cost sequence of nodes realized to reach node x_n . Each such pair defines a point in the chart depicted in Fig. 1 for a representative case example with over 2,500 kilometers. The continuous line in this figure is the lower bound envelope that optimistically estimates displacement and rest costs for different distances covered in a straight line from the origin. This lower bound envelope can be used to estimate $\hat{h}_{III}(x_n)$ because lowest costs realized until this moment in the search process, i.e., up to each location in the *permanent* subset, are a reasonable estimate of costs to be incurred going forward for the same distance in a straight line trajectory, and in particular until the final destination is reached.

This envelope is constructed progressively, beginning with the few rest stops which are in the vicinity of the origin of the trip, i.e., with low values of $dist(s, i_n)$, and continuing throughout the process, in a uncomplicated way. First, the domain of the straight line distance is divided at regular intervals: $[0, \Delta]$, $[\Delta, 2\Delta]$, $\dots$, $[(r-1)\Delta, r\Delta]$, $\dots$. Each included pair $[dist(s, i_n), \hat{g}(x_n)]$ corresponds to a point in the chart depicted in Fig. 1 and is associated to a specific interval index calculated as $r = \min\{k \in \mathbb{Z}_+ | k \geq \frac{dist(s, i_n)}{\Delta}\}$. For this interval update α_r as follows:

$$\alpha_r = \begin{cases} \frac{\hat{g}(x_n)}{dist(s, i_n)} & \text{if } \alpha_r \text{ is not defined} \\ \min \left[\alpha_r, \frac{\hat{g}(x_n)}{dist(s, i_n)} \right] & \text{otherwise} \end{cases} \quad (38)$$

These values of α_r are used to estimate $\hat{h}_{III}(x_n)$ as a function of the straight line distance of stoppage location i_n to the trip destination t , denoted by $\text{dist}(i_n, t)$. The corresponding interval index is now calculated as $r = \max\{k \in \mathbb{Z}_+ | k \leq \frac{\text{dist}(i_n, t)}{\Delta}\}$, and the heuristic is calculated as follows:

$$h_{III}(x_n) = \alpha_r \text{dist}(i_n, t) \quad (39)$$

4.2 Pruning the state-space graph

The physical road network consists of tens of thousands of road segments and stoppage locations (such as gas stations, rest areas, parking sites, road-side restaurants, etc.), each of which can be reached through many different sequences of predecessor stoppage points, with the result that any location can be reached at different times, characterizing different stoppage configurations in the space of states. Because the number of nodes grows exponentially and because we focus herein on real-time determination of low-cost routes, in addition to the heuristics, we prune the state-space graph to eliminate routing and scheduling alternatives which are less likely to result in reaching the final destination at minimum cost. This proposed pruning criterion is based on the concept of *weak dominance between nodes*.

Definition 1 Two nodes x_m and x_n are defined as similar in elements (i, p) , represented with the notation $x_m \stackrel{i,p}{\sim} x_n$, if and only if $i_m = i_n$ and $p_m = p_n$.

In other words, two stoppage configurations are similar in elements (i, p) , if the stop occurs in the same location and the type of the stop is the same, i.e., if the stop will be for rest, for meal, for overnight, or for weekend downtime in both stoppage configurations. Nodes that are similar in elements (i, p) likely will have different arrival times and different costs $\hat{g}(x)$ because incurred time is a continuous variable and any difference in elapsed time will lead to different costs to date.

Definition 2 x_m weakly dominates x_n , (or, analogously, x_n is weakly dominated by x_m) if and only if $x_m \stackrel{i,p}{\sim} x_n$ and $\hat{g}(x_m) < \hat{g}(x_n)$.

Proposition Consider the functions $\hat{h}_I(x)$, $\hat{h}_{II}(x)$ and $\hat{h}_{III}(x)$, presented in Sect. 5.1 and consider that $x_m \stackrel{i,p}{\sim} x_n$. Then, if x_m weakly dominates x_n , i.e., if $\hat{g}(x_m) < \hat{g}(x_n)$, there is no evidence that the lowest cost path between the origin and the destination configurations includes node x_n .

Demonstration: $\hat{h}_I(x_m) = \hat{h}_I(x_n) = 0$ always holds. If $x_m \stackrel{i,p}{\sim} x_n$ then $\hat{h}_{II}(x_m) = \hat{h}_{II}(x_n)$ and $\hat{h}_{III}(x_m) = \hat{h}_{III}(x_n)$ because $\hat{h}_{II}(x)$ and $\hat{h}_{III}(x)$ depend only on the straight line distance between the stoppage location associated with x . The estimates of the total costs of a path connecting the origin and final arrival configurations passing through nodes x_m and x_n are respectively $\hat{g}(x_m) = \hat{g}(x_m) + \hat{h}(x_m)$ and $\hat{f}(x_n) = \hat{g}(x_n) + \hat{h}(x_n)$. Given that $\hat{h}(x_m) = \hat{h}(x_n)$ and $\hat{g}(x_m) < \hat{g}(x_n)$ both hold it follows that $\hat{f}(x_m) < \hat{f}(x_n)$. Therefore there is no evidence that the path containing x_n is preferable to the path that contains x_m . $\square$

Pruning nodes in the graph is not without disadvantages as the guarantee of optimality for the problem is lost for both heuristics if the graph is pruned. However, the advantage is a significant improvement in processing speed, which allows the algorithm presented in the next section to be used in real time.

4.3 The algorithm

Let $s, t \in N$ be respectively the origin and destination of a trip and $x_s = (s, p_s, k_s, h_s^1, h_s^2, h_s^3, a_s, d_s, \delta_s)$ be the initial state or stoppage configuration at the origin, i.e., the location in which the trip originates, as described in Sect. 3. We now turn to finding a low cost path in the pruned graph of states connecting the origin to the destination by using the A^* algorithm with a node elimination process based on the weak dominance concept described in Sect. 4.2 as well as the proposed heuristics presented in Sect. 4.1 to estimate forward costs. The search algorithm consists of the following steps:

S1. *Initialization*: Set the initial node $x_s \leftarrow (s, \text{nil}, k_s, h_s^1, h_s^2, h_s^3, a_s, d_s, \delta_s)$ and destination location t . Let $\hat{g}(x_s) \leftarrow 0$, $\text{pred}(x_s) \leftarrow \text{nil}$ and $X_T = X_P = \emptyset$. Let $x_m \leftarrow x_s$. This initialization step attributes to the initial characteristics of the problem to the source node (location, nil stoppage type, initial travel day, respective initial driving times as zero, arrival and departure times as the same, and whether the main meal has already happened). Total accumulated costs are labeled zero, there is no predecessor, and both the temporary and the permanent lists are empty. At this point the initial node will be expanded so it is labeled as x_m .

S2. *Node expansion*: For each node $x_n \in \Gamma(x_m)$ do:

- a) if $\exists x_k \in X_P | x_k \stackrel{i,p}{\sim} x_n$, discard x_n , i.e., if there is a node in the permanent list that is similar in elements to x_n , discard x_n
- b) otherwise, calculate $\beta \leftarrow \hat{g}(x_m) + c(x_m, x_n)$, updating the cost and:
 - b.1) if $\exists x_k \in X_T | x_k \stackrel{i,p}{\sim} x_n$ and $\beta \geq \hat{g}(x_k)$, discard x_n , i.e., if the nodes are similar in elements and the cost associated to x_n is greater or equal, it is discarded;
 - b.2) if $\exists x_k \in X_T | x_k \stackrel{i,p}{\sim} x_n$ and $\beta < \hat{g}(x_k)$, i.e., if the nodes are similar in elements and the cost associated to x_n is smaller, set $X_T \leftarrow X_T \cup \{x_n\} - \{x_k\}$, i.e., update the temporary list by adding x_n ; and update x_n 's label letting $\hat{g}(x_n) \leftarrow \beta$ and $\text{pred}(x_n) \leftarrow x_m$, respectively the cost and the predecessor;
 - b.3) else (i.e., if the two nodes are not similar in elements) let $X_T \leftarrow X_T \cup \{x_n\}$ and update x_n 's label letting $\hat{g}(x_n) \leftarrow \beta$ and $\text{pred}(x_n) \leftarrow x_m$.

S3. *Node selection*: If $X_T = \emptyset$, i.e., if the temporary list is empty, there is no feasible sequence of stoppage configurations that satisfies all the conditions of the problem; exit with failure. Otherwise select a configuration $x_m \in X_T$ such that:

$$\hat{f}(x_m) = \min_{x \in X_T} \hat{f}(x) = \min_{x \in X_T} \hat{g}(x) + \hat{h}(x).$$

Let $X_T \leftarrow X_T - \{x_m\}$ and $X_P \leftarrow X_P \cup \{x_m\}$.

S4. Termination test: If for configuration x_m we have $i_m = t$, then finalize with success: the sequence of stoppages is determined by the predecessors of the configuration ending in x_m . Otherwise return to step S2.

In the above algorithm a change is needed when using the second heuristic, $\hat{h}_{III}(x_n)$. Before selecting a node, $\hat{h}_{III}(x)$ must be recalculated for all existing nodes $x \in X_T$ and after the selection, the heuristic function must be updated to include a new pair $[dist(s, i_m), \hat{g}(x_m)]$. In this case, step S3 is changed to:

S3. Node selection: If $X_T = \emptyset$, there is no feasible sequence of stoppage configurations that satisfies all the conditions of the problem; exit with failure. Otherwise recalculate $\hat{h}_{III}(x)$ for each $x \in X_T$ and select a configuration $x_m \in X_T$ such that:

$$\hat{f}(x_m) = \min_{x \in X_T} \hat{f}(x) = \min_{x \in X_T} \hat{g}(x) + \hat{h}_{III}(x)$$

Let $X_T \leftarrow X_T - \{x_m\}$, $X_P \leftarrow X_P \cup \{x_m\}$ and update the lower bound envelope function $\hat{h}_{III}(x)$ to include the pair $[dist(s, i_n), \hat{g}(x_n)]$.

Please see pseudo-code of the main procedures in the Appendix.

5 Numerical illustration

Brazil is the fifth country in the world in area, and despite its vast size much of the freight transportation is conducted over highways. It is not uncommon for a full load to be transported over long distances by one driver, who picks up and delivers the load in trips that may last several days. In such cases compliance to regulations becomes critical due to the added stress the driver is under: there are locations in the country where the support network for meals and resting is not yet well developed, and indeed in certain regions it is sparse, so availability besides costs are critical components in determining whether to stop and what route to choose, which directly affects load delivery costs and trucking profitability. Current Brazilian truck driver activities are governed by law n° 13.103/2015, approved on March 2, 2015. Among other provisions, this law caps the total number of driving hours in a week, establishes a minimum required stop in an appropriate rest area of thirty minutes for each continuous five-and-a-half hours of driving, requires at least one meal break per day lasting at least one hour, mandates an overnight resting time of at least eleven hours and a weekly rest period of at least thirty-five hours, and caps the number of overtime driving hours in a day.

This section presents computational tests to illustrate the use of the state-space shortest-path A^* algorithm described above to determine routes and stoppage schedules considering current Brazilian regulatory driver rest requirements. These illustrations use the highway grid in Brazil obtained from ANTT (Agência Nacional de Transportes Terrestres – the Brazilian ground transportation federal agency), and data on each municipality, with rest stops limited to county seats for illustrative purposes. Stops were scheduled using parameters from Brazilian regulations and prices observed in the market. We report on two sets of tests to analyze results as well as processing speeds. The first set involves examining one origin-destination pair for

sensitivity analyses: (a) a solution for the origin-destination pair is given and characteristics of the trip are analyzed; (b) changes in rest stop costs and services along the way are made and the altered result is explored further; (c) lastly, more comprehensive changes in costs in the region are made to conduct sensitivity analyses regarding the given route. A second set of tests for several origin-destination pairs is presented for heuristics comparison purposes, i.e., to examine the algorithm's performance with each of the heuristics described in the previous section.

5.1 An illustrative sensitivity analysis for one trip

This illustrative application involves a trip between the cities of Santa Maria in Rio Grande do Sul, the southernmost state in Brazil, and São José do Rio Preto in the state of São Paulo, for a total distance of approximately 1,440 kilometers (900 miles) depending on the routes taken. This sample trip was chosen because it represents a common trip from the rice-producing southern part of the country to Brazil's highest GDP state. The trajectory traverses a relatively developed region with many alternative routes and scheduling possibilities.

5.1.1 Base case (Case A)

Table 1 presents relevant results of the case study including parameters describing each stoppage configuration or node: location; type of stop (1, 2, 3, 4); arrival time; departure time; trip duration since the prior node; cumulative trip duration since departure from the origin (excluding stoppage time); distance traveled since the prior node; accumulated distance traveled from origin; cost since the prior node $c(x_m, x_n)$; cumulative cost since the origin $\hat{g}(x_i)$; and estimate of total trip cost at each node $\hat{f}(x_i)$.

As depicted in Table 1, total travel time between Santa Maria (RS) and São José do Rio Preto (SP) was 24 h and 42 min, covering a distance of 1,435 kilometers. This trip had four rest stops, two meal stops, and two overnight stops. The total estimated cost including truck relocation costs, stoppage costs, non-time-dependent service costs incurred during stops, driver labor costs, and opportunity costs, was BR\$ 4,951.82. In this example the driver rested a total of 26 hours for a total trip duration of 50 h and 42 min. Figure 2 - Case A depicts the chosen route.

5.1.2 Selected stoppage service cost change sensitivity analysis (Case B)

In Case A eight stoppage locations were chosen. Stoppage costs of selected service locations on Case A's chosen route were changed to verify model sensitivity. For this test meal costs were increased by BR \$100.00 and overnight costs were increased by BR\$ 150.00 for the four following locations: São Pedro das Missões, Cândói, Piraí do Sul e Cafelândia. The updated route after these increases in service costs is presented in Table 2, and Figure—Case B depicts the updated route. The substantial aggregate BR\$ 500.00 increase in costs in the four locations resulted in rerouting through locations with lower service costs. Although a substantial aggregate cost increase occurred in the Case A route, the rerouting resulted in a total cost impact of only BR\$ 68.94.

Table 1 Case A results

Location	Type	Arrive	Depart	Partial time	Total time	Partial km	Total km	$c(x_m, x_n)$	$\hat{g}(x_i)$	$\hat{f}(x_i)$
Santa Maria	4		07:00	0:00	0:00					
Julio de Castilhos	1	10:14	10:44	3:14	3:14	67	67	497.82	497.82	9,759.20
São Pedro das Missões*	2	13:46	14:46	3:01	6:16	191	258	508.13	1,005.95	4,440.36
Caibi	1	16:19	16:49	1:32	7:49	92	350	247.84	1,253.78	4,417.26
Candói*	3	20:38	07:38	3:49	11:38	243	593	1,108.40	2,362.18	4,767.76
Carambei	1	11:28	11:58	3:49	15:28	267	860	584.95	2,947.13	4,775.84
Pirai do Sul*	2	12:46	13:46	0:47	16:16	48	908	179.20	3,126.34	4,775.46
Sto Antonio da Platina	1	17:02	17:32	3:16	19:32	173	1,081	503.22	3,629.56	4,836.15
Cafelândia*	3	20:34	07:34	3:02	22:34	213	1,294	1,009.18	4,638.74	5,214.85
São José do Rio Preto	1	09:42		2:07	24:42	141	1,435	313.08	4,951.82	4,951.82

*These locations had their costs increased for Case B sensitivity analysis

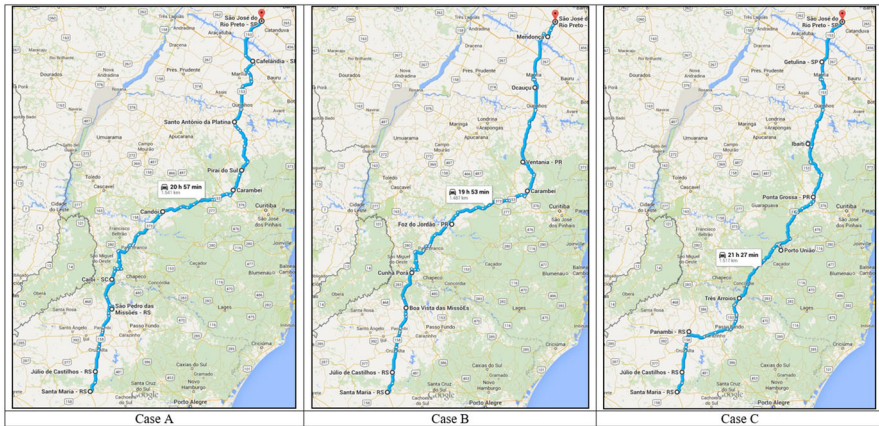


Fig. 2 Routing for each of the three cases (Source: Google Maps)

5.1.3 Widespread stoppage service cost change sensitivity analysis (Case C)

For Case C, widespread cost changes throughout the region were implemented as both meal and overnight costs were raised throughout the state of Paraná. The chosen route for this new scenario is depicted in Table 3, in which a significant routing change, to avoid higher service costs, can be observed. Although in every location in the state of Paraná, meal and overnight costs were raised respectively to BR\$ 100.00 and BR\$ 150.00, total trip costs were only BR \$ 194.02 higher than Case A, not a substantial increase. Figure 2—Case C shows the chosen route.

Had the Case A route been chosen, in which there were two service stops in the state of Paraná (one for a meal and one for an overnight) would have suffered a total impact of BR\$ 250.00. Therefore, despite the significant route change, the algorithm was able to find an alternative that offset the service cost increases applied to the whole state of Paraná. The route now suggests only one meal stop in that state and avoids the overnight stop suggested in Case A.

5.2 Comparison among heuristics

The three proposed functions $\hat{h}(x)$ were run on a set of twenty-eight sample trips for comparison purposes. These trips encompassed long-haul routes covering cities in each of Brazil's five macro-regions. The chosen cities were: Belém (PA), located in the Northern region on the mouth of the Amazon River; Recife (PE) and Salvador (BA), which are seaside cities located in the Northeastern region of the country; Pelotas (RS), located in the southern portion of the country's southernmost state; Campo Grande (MS) and Brasília (DF), located in the Mid-western region; and Niterói (RJ) and São Paulo (SP), which are located in the densely populated and most developed Southeastern region. For each origin-destination pair the three heuristic functions were run and trip costs, computer processing time, and number of temporary and permanent

Table 2 Case B results

Location	Type	Arrive	Depart	Partial time	Total time	Partial km	Total km	$c(x_m, x_n)$	$\hat{g}(x_i)$	$\hat{f}(x_i)$
Santa Maria	4		07:00	0:00	0:00					
Júlio de Castilhos	1	10:14	10:44	3:14	3:14	67	67	497.82	497.82	9,759.20
Boa Vista das Missões	2	13:51	14:51	3:06	6:21	195	262	528.21	1,026.03	4,429.27
Cunha Porã	1	16:50	17:20	1:58	8:20	117	379	315.71	1,341.73	4,421.72
Foz do Jordão	3	20:44	07:44	3:24	11:44	220	599	1,081.05	2,422.79	4,815.86
Carambeí	1	11:30	12:00	3:46	15:30	263	861	576.46	2,999.24	4,827.95
Ventania	2	13:46	14:46	1:45	17:16	100	961	322.93	3,322.18	4,890.13
Ocaçu	1	18:37	19:07	3:51	21:07	230	1,191	598.50	3,920.68	4,892.02
Mendonça	3	21:43	08:43	2:35	23:43	186	1,376	986.16	4,906.84	5,259.57
São Jose do Rio Preto	1	09:29		0:46	24:29	48	1,424	113.92	5,020.76	5,020.76

Table 3 Case C results

Location	Type	Arrive	Depart	Partial time	Total time	Partial km	Total km	$c(x_m, x_n)$	$\hat{g}(x_i)$	$\hat{f}(x_i)$
Santa Maria	4		07:00	0:00	0:00					
Júlio de Castilhos	1	10:14	10:44	3:14	3:14	67	67	497.82	497.82	9,759.20
Panambi	2	12:23	13:23	1:38	4:53	107	174	313.71	811.53	4,493.85
Três Arroios	1	16:49	17:19	3:26	8:19	228	402	530.49	1,342.02	4,487.44
Porto União	3	20:51	07:51	3:32	11:51	222	624	1,086.77	2,428.80	4,912.36
Ponta Grossa	2	11:49	13:00	3:57	15:49	214	837	749.42	3,178.22	5,088.20
Ibaiti	1	16:29	16:59	3:29	19:19	179	1,016	534.20	3,712.42	5,103.03
Getulina	3	20:50	07:50	3:50	23:09	261	1,277	1,140.65	4,853.07	5,485.62
São Jose do Rio Preto	1	09:49		1:59	25:09	133	1,410	292.77	5,145.84	5,145.84

nodes, respectively $|X_T|$ and $|X_P|$, were examined as depicted in Table 4. Computer processing times in the table correspond to an Intel Core i5-3337U 1,8 GHz processor.

As expected, $\hat{h}_I(x)$ provides the lowest cost result in every case. The proposed algorithm's null heuristic leads it to behave like Dijkstra's over the pruned graph. Heuristic $\hat{h}_{II}(x)$ proved to be very effective: all costs are equal to those obtained with function $\hat{h}_I(x)$, except for one trip (São Paulo to Brasília for which the difference was a 0.26% cost increase). Processing time was lower than $\hat{h}_I(x)$ in every case. Heuristic $\hat{h}_{III}(x)$ does not underestimate remaining costs with certainty due to graph pruning so it also cannot guarantee optimality over the pruned graph. On the other hand, heuristic $\hat{h}_{III}(x)$'s efficiency is observed in the shorter computer processing times. For 28,6% of the origin-destination pairs the results were the same as obtained by $\hat{h}_I(x)$. On average heuristic $\hat{h}_{III}(x)$ results were 0.63% worse than heuristic $\hat{h}_I(x)$ and in the worst case they were 2.20% worse, a very small difference especially considering that travel times are subject to significant variability. As expected, heuristic $\hat{h}_{III}(x)$ shows the lowest computer processing time.

Comparing computer processing speeds, heuristic $\hat{h}_{III}(x)$ is faster than the other two as function $\hat{h}_I(x)$ is on average 14.6 times slower and heuristic $\hat{h}_{II}(x)$ (also on average) is 3.7 times slower. In extreme cases, heuristic $\hat{h}_{III}(x)$ was sixty times faster than $\hat{h}_I(x)$ and sixteen times faster than heuristic $\hat{h}_{II}(x)$.

Figure 3 shows three templates depicting the relationship between computer processing times and the number on nodes in the temporary sets X_T for each of the three methods. In every instance a polynomial increase in processing time relative to the number of nodes is apparent. However, the rate of growth is lowest for function $\hat{h}_{III}(x)$ and highest for function $\hat{h}_I(x)$. When comparing computer processing times with the number on nodes in the permanent sets X_P , the three heuristics seem to behave in similar fashion, as visually represented in Fig. 4. However heuristic $\hat{h}_{III}(x)$ includes a substantially lower number on nodes in X_P as compared to the other heuristics.

Of all tests performed, only four times did heuristic $\hat{h}_{III}(x)$ use more than ten seconds of CPU time. These instances are listed in Table 5. An additional test was performed for each in an attempt to lower computer processing time. This additional test consisted of a change in heuristic $\hat{h}_{III}(x)$ to add a certain percentage of expected costs to the lower bound envelope. Results show that as this percentage grows, CPU time diminishes exponentially tending asymptotically to zero, as depicted in Fig. 5. On the other hand, an increase in routing and scheduling costs is observed, as these decrease by 0.13% for each percentage point added to the heuristic, as can be observed in the chart in Fig. 6.

These computational results are an indication that the methodology proposed herein could be helpful in finding a low cost combination of vehicle routing and driver scheduling in shorter processing times, potentially allowing for its broader use in in-vehicle routing guidance systems.

6 Conclusion

This paper addresses the long-haul point-to-point vehicle routing and driver scheduling problem subject to hours-of-service regulatory constraints, which is especially relevant

Table 4 Heuristic function comparisons

From	To	Function $\hat{h}_I(x_n)$, null heuristic			Function $\hat{h}_{II}(x_n)$			Function $\hat{h}_{III}(x_n)$					
		Cost	CPU time	$ x_T $	$ x_P $	Cost	CPU time	$ x_T $	$ x_P $	Cost	CPU time	$ x_T $	$ x_P $
Belém	Salvador	5,779.25	3.84	5,727	3,216	5,779.25	0.50	2,57	1,074	5,807.41	0.15	1,386	277
	Pelotas	13,417.99	60.50	11,112	10,231	13,417.99	32.49	9,901	6,649	13,675.53	7.48	7,063	3,274
Belém	Cp. Grande	10,676.16	7.86	4,535	3,622	10,676.16	3.60	3,316	2,889	10,733.80	3.43	3,201	2,456
Belém	Niterói	10,847.52	52.51	12,825	11,082	10,847.52	21.67	10,666	7,883	11,011.40	11.95	9,17	5,356
Belém	São Paulo	8,557.46	66.14	14,089	12,087	8,557.46	28.53	9,622	7,876	8,557.46	6.92	5,919	2,444
Belém	Brasília	6,067.17	0.97	3,084	2,101	6,067.17	0.36	1,954	1,019	6,089.71	0.03	673	110
Recife	Belém	5,676.87	10.85	6,432	6,015	5,676.87	7.10	5,565	3,737	5,676.87	2.54	4,17	1,531
Recife	Salvador	1,894.16	1.18	2,156	1,151	1,894.16	0.47	1,477	498	1,894.16	0.14	1,188	150
Recife	Pelotas	11,788.83	150.32	16,217	14,424	11,788.83	92.27	14,129	11,611	11,847.51	22.54	10,383	4,375
Recife	Cp. Grande	10,652.24	42.33	13,119	10,905	10,652.24	50.04	10,912	9,629	10,746.87	24.05	9,79	5,645
Recife	Niterói	6,316.23	8.03	6,859	4,949	6,316.23	4.03	5,331	2,972	6,370.25	1.50	4,066	1,038
Recife	São Paulo	7,319.27	26.72	10,421	8,201	7,319.27	13.97	8,119	5,871	7,416.50	5.33	6,122	2,166
Recife	Brasília	6,054.00	9.16	8,051	6,181	6,054.00	5.61	5,226	3,838	6,054.00	2.33	4,193	1,769
Salvador	Pelotas	10,174.92	108.92	12,768	12,371	10,174.92	83.92	12,321	10,892	10,332.69	41.17	10,6	6,357
Salvador	Cp. Grande	7,610.79	76.27	10,934	9,24	7,610.79	30.05	9,136	5,83	7,650.07	1.93	3,737	1,047
Salvador	Niterói	4,750.19	4.26	4,394	2,801	4,750.19	1.25	3,541	1,418	4,821.55	0.50	2,12	530
Salvador	São Paulo	5,330.09	11.10	6,841	4,262	5,330.09	1.94	4,703	1,749	5,334.70	0.39	2,274	416
Salvador	Brasília	4,066.54	1.14	3,378	2,051	4,066.54	0.34	1,782	926	4,066.54	0.20	1,479	511
Pelotas	Cp. Grande	5,320.53	35.98	7,082	6,001	5,320.53	12.84	6,114	4,023	5,320.53	4.66	4,704	2,078

Table 4 continued

From	To	Function $\hat{h}_I(x_n)$, null heuristic				Function $\hat{h}_{II}(x_n)$				Function $\hat{h}_{III}(x_n)$			
		Cost	CPU time	$ x_T $	$ x_P $	Cost	CPU time	$ x_T $	$ x_P $	Cost	CPU time	$ x_T $	$ x_P $
Pelotas	Niterói	4,188.31	12.29	6,217	3,922	4,188.31	2.04	3,64	1,274	4,280.65	0.54	2,341	496
Pelotas	São Paulo	3,432.86	13.07	5,744	4,479	3,432.86	2.74	4,122	1,306	3,433.66	0.92	2,767	578
Pelotas	Brasília	5,984.53	82.49	10,391	8,718	5,984.53	22.73	8,233	4,105	6,066.92	2.16	4,448	1,127
Cp. Grande	Niterói	3,453.32	19.36	7,422	4,448	3,453.32	1.94	3,555	725	3,496.28	0.92	2,793	506
Cp. Grande	São Paulo	2,608.20	12.03	5,465	3,442	2,608.20	5.12	3,617	1,463	2,614.72	4.48	3,506	1,324
Cp. Grande	Brasília	3,468.53	4.16	3,71	2,126	3,468.53	0.56	1,947	556	3,468.53	0.58	1,575	371
Niterói	São Paulo	866.11	0.60	1,827	313	866.11	0.03	497	17	868.35	0.01	388	2
Niterói	Brasília	3,276.41	10.85	5,275	3,25	3,276.41	0.89	3,71	736	3,276.41	0.36	2,484	419
São Paulo	Brasília	2,733.18	16.48	6,022	4,281	2,740.20	6.08	4,368	1,821	2,740.20	4.47	4,029	1,463
Average		6,153.99	30.34	7,587	5,899	6,154.24	15.47	6,006	3,576	6,201.90	5.42	3,509	1,708

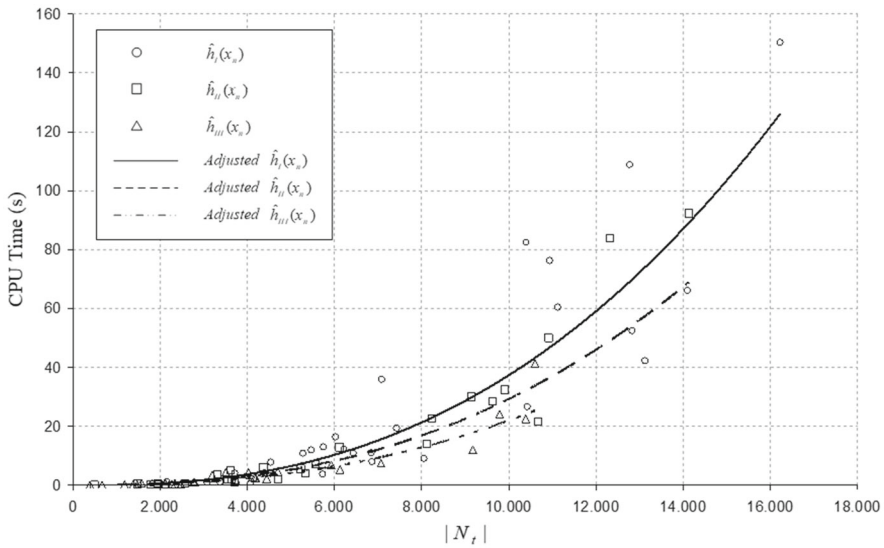


Fig. 3 CPU Time $\times$ number of nodes in the temporary set for each of the three heuristics

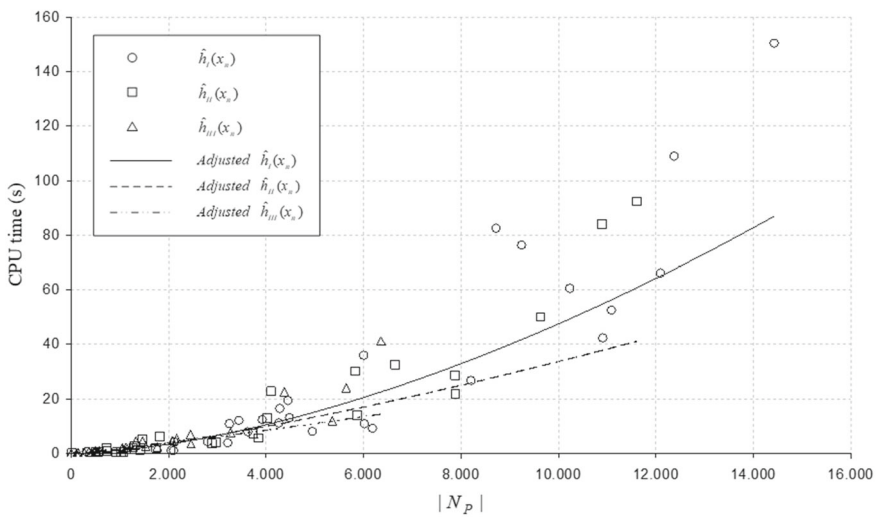


Fig. 4 CPU Time $\times$ number of nodes in the permanent set for each of the three heuristics

in countries with large land area in which highway freight is significant. In recent decades most such countries have enacted truck driver hours-of-service regulations to improve working conditions and increase road safety by limiting hours of continuous driving and by requiring meal, overnight, and weekly rests of minimum duration. This more rigorous regulatory environment together with the proliferation of real-time in-vehicle routing tools has led to the desirability of models and algorithms that quickly provide cost-effective alternative routes taking into account available stoppage

infrastructure whenever drivers face unexpected conditions mid-travel. This paper addresses the problem by searching for low-cost paths in a dynamically expanded state-space graph with an adapted A^* algorithm. In addition, the paper illustrates use of the methodology in the Brazilian regulatory environment and examines the computational efficiency of the algorithmic solution.

The tradeoffs between routing costs and required stoppage costs are modeled as a graph in a multi-dimensional state-space in which individual nodes with spatial, temporal, and stoppage category characteristics are expanded sequentially. We did not find in the literature a treatment of the long-haul VRTDS-HOS problem in which times of arrival and departure in every stop along the way were treated as continuous variables, therefore allowing for an examination of route alternatives within a shortest-path framework in a state-space with both temporal and spatial characteristics. Besides applying the model to the Brazilian regulatory environment, the paper also contributes to the methodological academic literature on the VRDSP-HOS by using an adaptation of the label-setting A^* algorithm to find the low-cost path in the aforementioned iteratively built state-space graph. The paper proposes two heuristics for the adapted A^* algorithm, one of which “learns” from information acquired during the trajectory from the origin through each node expansion, i.e., it uses cost data obtained as the trip evolves to narrow the number of feasible successor nodes and therefore decrease processing time. Significantly shorter processing times were achieved using these heuristics.

This research has several managerial implications. First, the joint consideration of temporal, spatial, and stoppage attribute variables is desirable when long-haul trips are planned, as different routes can have very different stoppage costs, affecting profitability. Second, besides helping managers, the potential use of the model by truck drivers in real-time, i.e., during trips, provides them with a desirable level of autonomy, allowing for in-trip route adjustments to adapt to changing conditions. Indeed, because we feel this approach and framework are helpful to managers, we hope there may be other opportunities to model certain types of multi-attribute vehicle routing and driver scheduling problems by broadening the optimization state-space beyond geographic location parameters to simultaneously also include temporal and stoppage attribute dimensions and by finding least-cost paths in graphs representing those respectively expanded state spaces.

Table 5 Results for different percentage increases in $\hat{h}_{III}(x)$

Route	From	To	$\hat{h}_{III}(x_n)(+0\%)$		$\hat{h}_{III}(x_n)(+12\%)$		$\hat{h}_{III}(x_n)(+25\%)$		$\hat{h}_{III}(x_n)(+50\%)$	
			Cost	CPU time	Cost	CPU time	Cost	CPU time	Cost	CPU time
R1	Belém	Niterói	11,011	11.95	11,021	7.89	11,054	2.66	11,054	1.44
R2	Recife	Pelotas	11,847	22.54	11,975	14.02	12,707	12.62	13,533	1.74
R3	Recife	Campo Grande	10,746	76.13	10,897	50.24	10,897	28.43	10,897	8.89
R4	Salvador	Pelotas	10,332	41.17	10,374	19.24	10,729	3.08	11,197	0.91
Average increase over $\hat{h}_I(x_n)$			1.11%		1.86%		4.36%		7.26%	

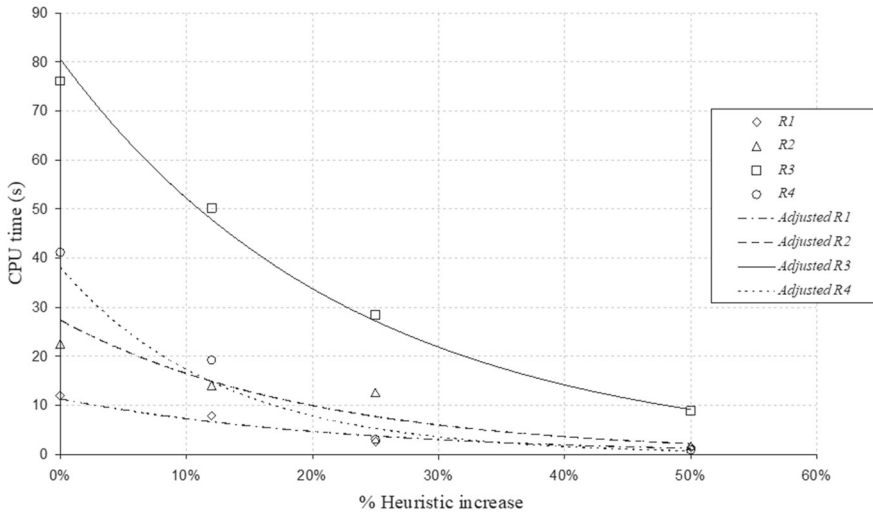


Fig. 5 $\hat{h}_{III}(x_n)$ performance as a function of the percentage increase

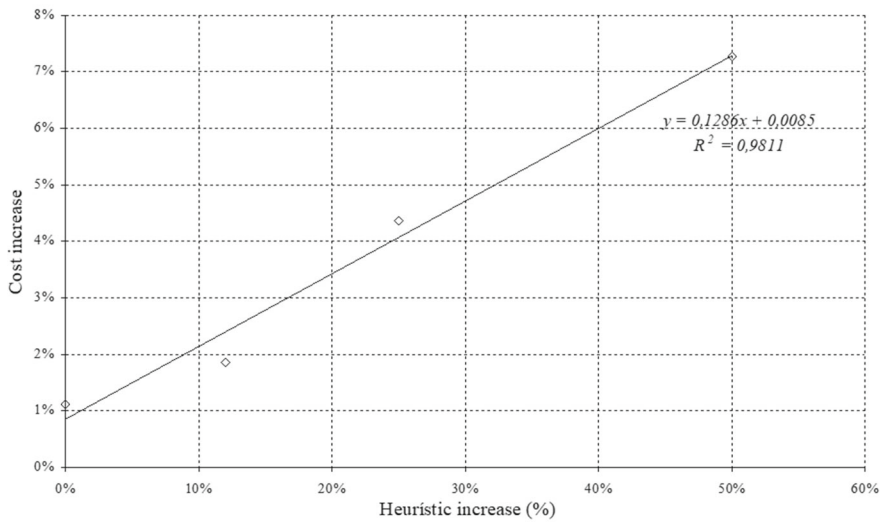


Fig. 6 Trip cost percentage increase as function of $\hat{h}_{III}(x_n)$ changes

A Appendix

Algorithm pseudo-code for the computer system as programmed

Procedure FindSolution(Origin, Destination);

Begin

 Data Input;

 CalculateDistances (Origin, Destination);

```

AStop ← CreateStop(Origin,Type=S,Pred=nil);
TempList.Include(AStop)
While (AStop < > nil) and (AStop.Location < > LDestination) do
    TempList.Remove(AStop);
    PermList.Include(AStop);
    UpdateEnvelop(AStop);
    TempList.Include(Successors(AStop));
    AStop ← FindBestStop;
End;
End;
Return AStop;
End.

Procedure CreateStop(ALocation,AType,APred);
Begin
    NewStop.Pred ← APred;
    NewStop.Type ← AType;
    NewStop.Location ← ALocation;
    if APred = nil then do
        NewStop.HC ← 0;
        NewStop.HS ← MinTimeStartWorkday;
        NewStop.h1 ← 0;
        NewStop.h2 ← 0;
        NewStop.h3 ← 0;
        NewStop.gHat ← 0;
    End do;
Else
    TravelTime ← HighwayNetwork.Elapsed.Time(APred.Location,ALocation)
    NewStop.HC ← APred.HS + TravelTime;
    NewStop.NoMeal ← True;
    if
        AType = Rest then Begin
            NewStop.HS ← NewStop.HC + RestT;
            NewStop.NoMeal ← Pred.NoMeal;
        End;
        AType = Meal then Begin
            NewStop.HS ← NewStop.HC + MealT;
            NewStop.NoMeal ← False;
        End;
        AType = Night then NewStop.HS ← NewStop.HC + NightT;
        AType = Week then NewStop.HS ← NewStop.HC + WeekT;
    End if;
    NewStop.h1 ← TravelTime;
    NewStop.h2 ← TravelTime;
    NewStop.h3 ← TravelTime;
    if APred.Type in [Rest,Meal] then
        NewStop.h2 ← APred.h2 + TravelTime;
    End if;
    if APred.Type in [Rest,Meal,Night] then NewStop.h3 ← APred.h3 + TravelTime;
    End if;
    DisplCost ← HrTruckCost * TravelTime + DriverCost(APred,TravelTime);
    StopCost ← (NewStop.HS - NewStop.HC) * HrStopCost (ALocation) +
    ServiceCost(ALocation,Type);

```

```

NewStop.gHat ← APred.gHat + DisplCost + StopCost;
if (NewStop.h3 > WkLimt) or (NewStop.h2 > DayLimt + Ovrtime) then NewStop.gHat ← large;
End if;
if Pred.NoMeal then
  If Meal(NewStop.HC)
    During: if AType <> Meal then NewStop.gHat ← Large;
  End if;
  After: NewStop.gHat ← Large;
  End if;
End if;
Return NewStop;
End.

```

```

Procedure Meal(H);
Begin
  Day ← Trunc(H/24);
  MBegin ← 24*Day + MealBegin;
  MEnd ← 24*Day + MealEnd;
  If H < MBegin then
    Return Before;
  If H > MEnd then
    Return After;
  Else
    Return During;
  End If;
End.

```

```

Procedure UpdateEnvelope(AStop);
Begin
  k ← Trunc(AStop.Location.DistOrigin/100);
  Theta ← AStop.gHat/AStop.Location.DistOrigin;
  If Theta < H(k) then
    H(k) ← Theta;
  End if;
End.

```

```

Procedure fHat(AStop);
Begin
  k ← Trunc(AStop.Location.DistDestin / 100);
  hHat ← AStop.Location.DistDestin * H(k);
  fHat ← hHat + AStop.gHat;
  Return fHat;
End.

```

```

Procedure FindBestStop;
Begin
  BestNode ← TemporaryList.First;

```

```

For CurrentNode in TemporaryList do
  If fHat(CurrentNode) < fHat(BestNode) then
    BestNode ← CurrentNode;
  End do;
End For;
Return BestNode;
End.

```

```

Procedure Successors(AStop);
Begin
  AuxList ← HighwayNetwork.NeighborhoodLocationSearch(AStop);
  For Location in AuxList do
    For Type in TypeStop do
      NewStop ← CreateStop(Location, Type, AStop);
      TemporaryList.Include(NewStop);
    End do;
  End do;
End.

```

```

Procedure DriverCost(APred, TravelTime);
Begin
  Total ← 0;
  If Pred.Stop in [Rest, Meal] then Total Pred.h2 + Travelttime;
  End if;
  If Pred.Stop in [Night, Week] then Total ← TravelTime;
  End if;
  If Total ≤ Workday then
    RegularTime ← TravelTime;
    OverTime ← 0;
  Else do
    OverTimes ← Total — WorkDay;
    RegularTime ← WorkDay;
  End if;
  End do;
  DriverCost ← RegularTime * RegularPay +
    OverTime * OverTimePay;
  Return DriverCost;
End.

```

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